

Contrastive Satellite Representation Learning for Commodity Futures Prediction

1. What is the problem you will be investigating? Why is it interesting?

We investigate whether self-supervised visual representations learned from satellite imagery can predict agricultural commodity futures returns (specifically corn, soybean, and wheat front-month contracts). Traditional commodity forecasting relies on USDA crop reports released with substantial delay (monthly, with revisions). Satellite-derived vegetation indices (e.g., NDVI) are known predictors of crop yields, but existing approaches use hand-crafted indices as point features, discarding the rich spatial structure of the imagery.

The core question is: **can a contrastive learning framework extract latent "crop health" representations from raw satellite imagery that carry predictive signal for futures prices, beyond what simple vegetation indices provide?** This is interesting because (a) it tests whether deep visual representations capture information about agricultural supply shocks that is not yet priced into markets, (b) it connects the remote sensing and financial economics literatures in a novel way, and (c) it addresses a real information asymmetry - satellite imagery is available days before official reports.

2. What dataset(s) are you using? Is that sufficient?

- **Satellite data:** Sentinel-2 L2A multispectral imagery (4-band: RGB+NIR, 10m resolution) from Google Earth Engine, covering the U.S. Corn Belt (Iowa, Illinois, Indiana, Nebraska, Minnesota) from 2017 to 2025. We construct **monthly cloud-free composites** per county (z meters x z meters of farmland).
- **Financial data:** CME front-month futures daily settlement prices for corn (ZC), soybeans (ZS), and wheat (ZW), aggregated to monthly returns. Source: Yahoo Finance or Quandl/Nasdaq Data Link.
- **Auxiliary labels (for evaluation only, not training):** USDA NASS county-level crop yield estimates (annual), used as a sanity check for whether the learned representations encode yield information.

Sufficiency: 310K image patches provide ample data for contrastive pretraining. The financial target is limited (~100 monthly return observations per commodity), which is a known challenge in financial prediction. We mitigate this via cross-commodity and cross-region pooling, and via the expanding-window protocol from Gu et al. (2020).

3. What deep learning approach will you use/develop?

We propose **Geo-SimCLR**, a domain-adapted contrastive learning framework for satellite time series:

Phase 1 - Self-supervised pretraining (novel augmentation strategy): We adapt SimCLR (Chen et al., 2020) with **geospatially-informed augmentations**. Standard SimCLR augmentations (color jitter, random crop) are inappropriate for multispectral satellite imagery. Our novel augmentation set includes:

- **Spectral band dropout:** randomly zeroing one of the 4 input bands (forces the encoder to not rely on any single band).
- **Temporal positive pairs:** two cloud-free images of the *same* county from adjacent months serve as positive pairs (the landscape changes slowly, so same-location across short time = semantic similarity). This is the key novelty - standard SimCLR uses augmented copies of the same image, whereas we use **temporal invariance as the self-supervisory signal**.
- **Geographic negative sampling:** negative pairs are drawn from geographically distant counties in different agro-climatic zones to ensure the contrastive loss separates genuinely different crop conditions.

The encoder is a ResNet-18 backbone producing a 512-d representation per image patch. The contrastive loss is NT-Xent.

Phase 2 - Downstream prediction: The frozen encoder produces a 512-d monthly feature vector per county. We aggregate county representations by spatial averaging across the Corn Belt to produce a single 512-d "crop state" vector per month. This vector is fed into a small temporal model (2-layer LSTM or linear probe) to predict next-month commodity futures returns.

Novelty: (a) temporal positive pairs as a self-supervisory signal for satellite data (not standard augmentation), (b) geographically-structured negative sampling, (c) the entire pipeline from raw pixels to financial return prediction, bridging two literatures.

4. What are anticipated challenges?

- **Cloud contamination:** Even with the L2A cloud mask, residual haze/thin clouds corrupt composites. Mitigation: use median compositing over each month's available scenes and exclude months with <3 clear observations.
- **Signal-to-noise ratio in financial targets:** Monthly commodity returns are extremely noisy (OOS R^2 for stock returns is typically 1-2%; commodities may be similar). The contrastive representations may capture real crop variation but the link to *futures prices* could be weak because markets partially anticipate crop conditions. We will test whether the signal concentrates during **surprise periods** (months where USDA reports deviate from consensus).

- **Compute:** SimCLR pretraining on 310K multi-spectral images is feasible on a single A100 GPU (comparable to ImageNet-scale pretraining on a subset). We budget ~24 hours for pretraining.
- **Representation collapse:** Contrastive learning can collapse to trivial solutions. We monitor representation entropy and use standard remedies (large batch size, projection head, temperature tuning).

5. How will you evaluate your results?

Quantitative:

- **OOS R^2** (Gu et al. 2020 definition, without demeaning the denominator) of predicted vs. realized commodity futures returns, using an expanding window (train 2017-2020, validate 2021, test 2022+, then roll forward).
- **Directional accuracy:** fraction of months where predicted return sign matches realized sign (>50% is informative).
- **Portfolio Sharpe Ratio:** a simple long/short strategy across the three commodities based on predicted return ranking.
- **Linear probe accuracy** on USDA yield quintiles (does the representation encode crop condition quality?).
- Compare against baselines: (a) raw NDVI county average as predictor, (b) a supervised CNN trained end-to-end on returns, (c) random features.

Qualitative:

- **Spatial attention maps** via GradCAM on the ResNet encoder, showing which image regions drive the representation (expect: active cropland, not urban/water).
- **t-SNE / UMAP visualization** of the learned 512-d representations colored by (a) month-of-year, (b) NDVI quintile, (c) subsequent commodity return sign. We expect temporal and crop-health clustering to emerge despite never training on labels.
- **Temporal representation trajectories:** plotting the first 2 principal components of the county-average representation over time, overlaid with commodity price, to visually assess whether representation dynamics lead price dynamics.